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27 June 1975
SUPERSEDING
MIL-E-1/1676
10 July 1969
(See note 3)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPES 2J30 THROUGH 2J34

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, fixed frequency, external magnet, rated peak power output 35 kw

ABSOLUTE RATINGS:

Parameter:	Ef	If	tk	tpc	Du	epy	ib	pi	Pi	T(anode)	VSWR	Alt
Unit:	V	A	sec	µs	---	kv	a	kw	W	°C	---	ft
Maximum:	6.93	4.0	---	2.50	0.002	22	30	660	600	100	1.5:1	10,000
Minimum	---	---	120	---	---	12	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Cathode: Unipotential (oxide coated)
Weight: 2 pounds, 4 ounces (approx)

TEST CONDITIONS:

Parameter:	Ef	tk	tpc	Du	Ib	VSWR	H
Unit:	V	sec	µs	---	mAdc	---	Gauss
Test condition 1:							
Maximum:	---	---	1.10	---	---	1.15	---
:	6.3	---	---	0.001	30	---	1,900
Minimum:	---	120	0.90	---	---	---	---
Test condition 2:							
Maximum:	---	---	1.10	---	---	1.15	---
:	6.3	---	---	0.001	12.5	---	1,200
Minimum:	---	120	0.90	---	---	---	---
Test condition 3:							
Maximum:	---	---	1.10	---	---	1.15	---
:	6.3	---	---	0.002	25	---	1,200
Minimum:	---	120	---	---	---	---	--- (Note 1)

Frequency		
Type	MHz	±
2J30	2,880	20
2J31	2,840	20
2J32	2,800	20
2J33	2,760	20
2J34	2,720	20

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TESTS	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4027	Temperature coefficient	-	1		ΔF/ΔT	---	0.07	MHz/°C
1047	Low-temperature operation	-	1	tk = 180 sec (max)	Po	240	---	W
1031	High-frequency vibration	-	-	No voltages applied	---	---	---	---
	Quality conformance inspection, part 1	2	-					
4289	Heater current	-	-	Ef = 6.3 V; tk = 120 sec (min)	If	1.35	1.65	A
4003	Pressurizing	-	-	40 to 45 psia	---	---	---	---
4306	Pulse voltage (1)	-	1		epy	18	22	kv
4250	Power output (1)	-	1	t = 300 (max)	Po	240	---	W
4308	RF bandwidth (1)	-	1		BW	---	2.5/tpc	MHz
---	Fixed frequency	-	1	2J30 2J31 2J32 2J33 2J34	F F F F F	2,860 2,820 2,780 2,740 2,700	2,900 2,860 2,820 2,780 2,740	MHz MHz MHz MHz MHz
4310	Frequency pulling figure	-	1		ΔF	---	15	MHz
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	-	-	No voltages applied	---	---	---	---
4309	Magnetron non-oscillating impedance	-	-					
	Standing-wave ratio	-	-		σ'	10:1	---	---
	Position of maximum	-	-	2J30 2J31 2J32 2J33 2J34	Distance Distance Distance Distance Distance	12.2 12.5 12.8 13.1 13.4	13.5 13.8 14.2 14.5 14.8	cm cm cm cm cm
4306	Pulse voltage (2)	-	2,3		epy	9.5	11.5	kv
4250	Power output (2)	-	2 3	t = 300 (max) t = 300 (max)	Po Po	35 70	--- ---	W W
4308	RF bandwidth (2)	-	2,3		BW	---	2.5/tpc	MHz
	Quality conformance inspection, part 3							
4551	Life test	-	1	Group D	t	500	---	hrs
---	Life-test end points:							
4250	Power output	-	1		Po	192	---	W
4308	RF bandwidth	-	1		BW	---	2.5/tpc	MHz

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NOTES:

1. Coil No. 400, or equivalent; pole tip No. 22-1.

2. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

3. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 80
DSA - ES

User activities:
Army - ME, AV
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2973)

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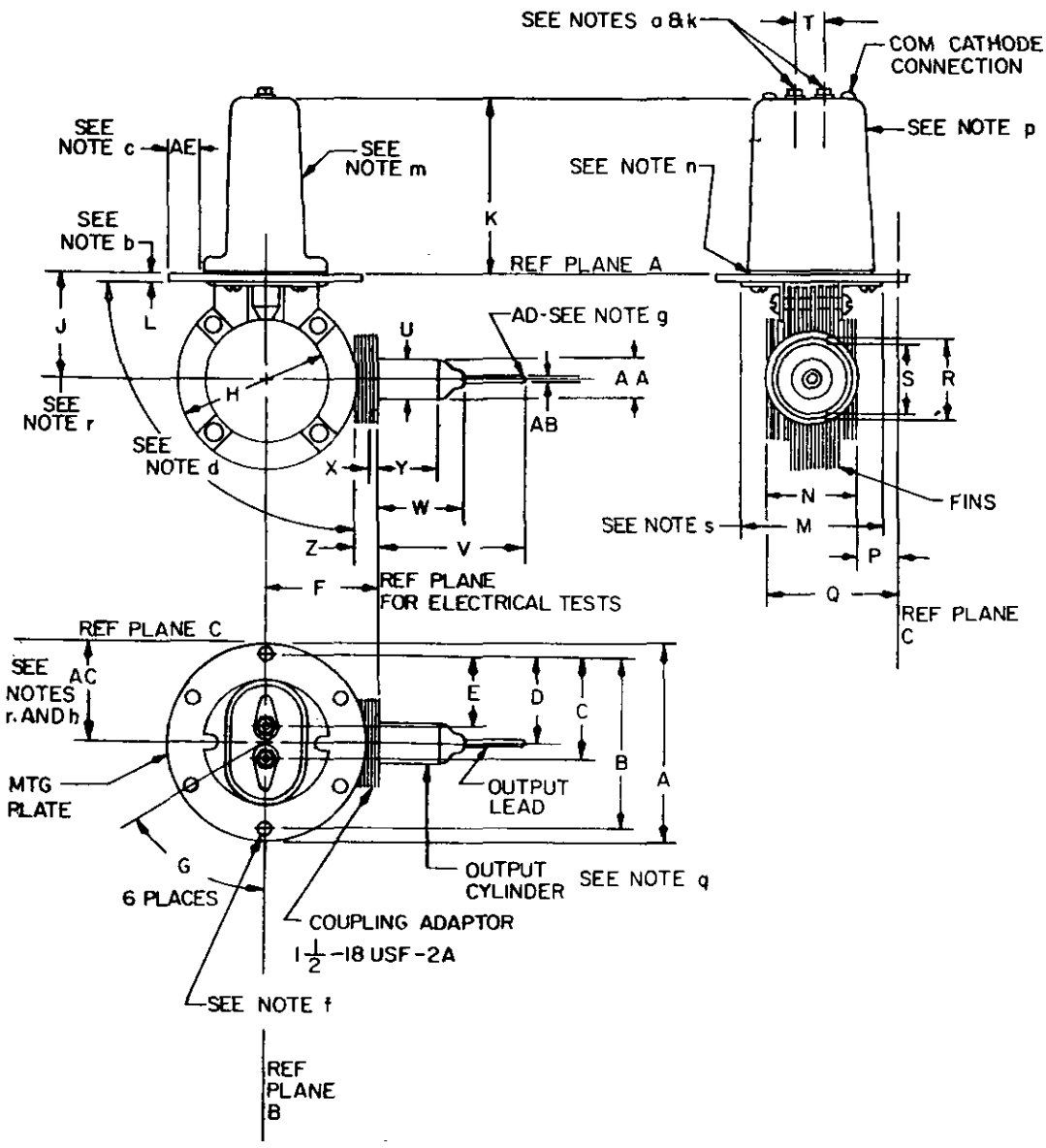


FIGURE 1. Outline drawing of electron tube types 2J30 through 2J34.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:16Z
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
F	1.828 (46.43)	1.922 (48.82)
J	1.677 (42.60)	1.697 (43.10)
R	1.365 (34.67)	1.385 (35.18)
S	1.117 (28.37)	1.127 (28.63)
T	.490 (12.45)	.510 (12.95)
U	.625 (15.88)	.670 (17.02)
V	2.375 (60.33)	2.500 (63.50)
W	1.375 (34.93)	1.438 (36.53)
X	.115 (2.92)	.135 (3.43)
Z	.365 (9.27)	.375 (9.53)
AB	.120 (3.05)	.127 (3.23)
AC	1.427 (36.25)	1.447 (36.75)
Quality conformance inspection, part 2		
A	3.219 (81.76)	3.281 (83.34)
B	2.869 (72.07)	2.881 (73.18)
G	59° 48'	60° 12'
H	2.938 (74.36)	3.063 (77.80)
K	2.922 (74.22)	3.047 (77.39)
M		2.480 (62.99)
N		1.490 (37.85)
P	.677 (17.20)	
Q		2.197 (55.80)
AA		.670 (17.02)
Reference dimensions		
C	1.687 (42.85)	
D	1.437 (36.50)	
E	1.187 (30.15)	
L	.125 (3.18)	
Y	1.000 (25.40)	
AD	.062R (1.57)	

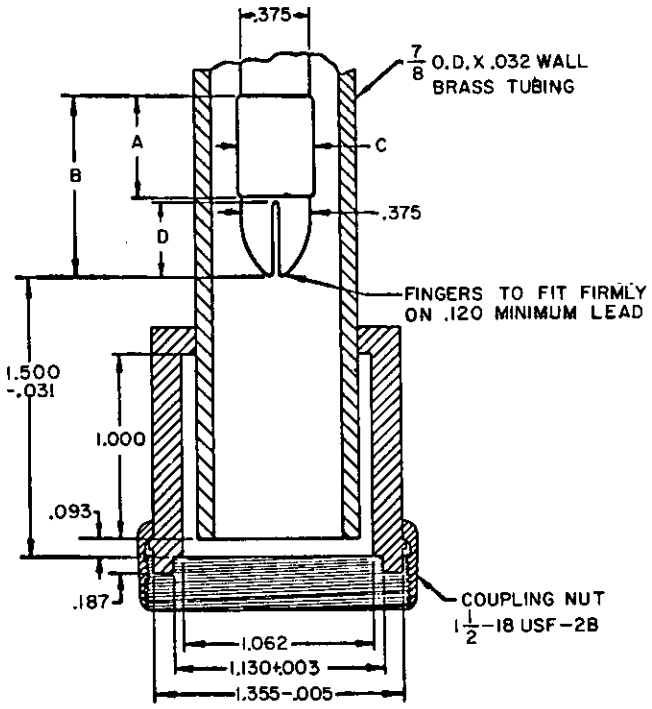
FIGURE 1. Outline drawing of electron tube types 2J30 through 2J34 - Continued.

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NOTES:
a. The jack holes shall be within a radius of .023 (.58) of the location specified but shall be spaced .500±.010 (12.70±.25) with respect to each other. The centerlines of the holes shall be perpendicular to reference plane 'A' within 3°.
b. Any portion of the assembly extending above this surface shall be within 1.109 (28.17) radius of the true center of the plate.
c. With the flange resting on a plane surface, the flatness of the mounting plate .500 (12.70) from the outer edge shall be such that a .010 (.25) thickness gage .125 (3.18) wide shall not enter more than .250 (6.35).
d. Areas between these planes and including these planes shall be painted with a heat resistant non-corrosive paint.
e. Soldered joints in the mounting flange and coupling adapter shall be vacuum tight so that the mounting flange may be used to provide a hermetic seal.
f. Drill .193±.003 (4.90±.08), 6 holes.
g. Centerline of lead measured at this end shall be concentric with centerline of coupling adapter within .020 (.51).
h. Centerline of output cylinder shall be concentric with centerline of coupling adapter within .010 (.25).
j. This surface of coupling adapter shall be parallel to reference plane 'B' within 1°.
k. Hex head banana pin jack .406 (10.31) long, hole .169±.005 (4.29±.13). Leads shall be flexible and slack.
m. Pyrex glass or porcelain or approved equivalent.
n. Gasket may be omitted by cementing the base to the plate.
p. Indicate common cathode connection by letter 'C' on this surface.
q. Output cylinder shall be non-corrosive material or shall be painted black, pated minimum 20 msi silver or minimum 10 msi gold.
r. This dimension applies to coupling adapter.
s. Number of cooling fins shall be 9, 10, or 11.

FIGURE 1. Outline drawing of electron tube types 2J30 through 2J34 - Continued.

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INCHES	MM	INCHES	MM
.003	.08	.375	9.53
.005	.13	1.000	25.40
.031	.79	1.062	26.97
.032	.81	1.130	28.70
.093	2.36	1.355	34.42
.120	3.05	1.500	38.10
.187	4.75		

Dim	Minimum	Maximum
C	.443 (11.25)	.445 (11.30)
Nominal dimensions		
A	1.052 (26.72)	
B	1.812 (46.02)	
D	.760 (11.28)	

FIGURE 2. Output coupling.

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